APEX: Automated Pipeline for Explorative Drug Discovery

A target-agnostic QSAR screening and molecular docking platform

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Computational Drug Discovery

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Outline

The Drug Discovery Challenge

Scale of the Problem

- 12–15 years from target to approval
- \$2.6 billion average per approved drug
- $\sim$1 in 10,000 screened compounds reaches trials

Computational Acceleration

Early-stage virtual screening reduces the wet-lab search space from millions to tens of candidates before any synthesis is attempted.

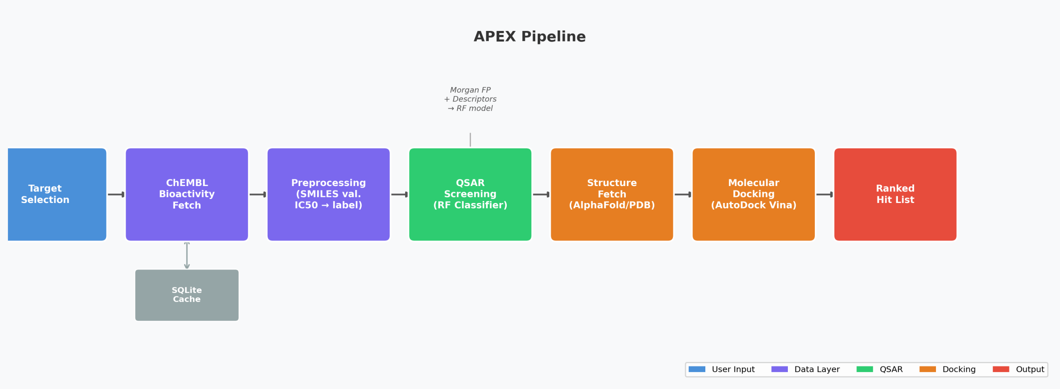
Bottleneck

Manual, target-specific pipelines require expert setup for every new biological target — not scalable.

Our Approach

APEX automates the full pipeline: data ingestion $\rightarrow$ QSAR $\rightarrow$ docking for *any* ChEMBL target, on demand.

APEX End-to-End Workflow



Each stage is independently cached and testable; the web platform exposes every stage through a REST API.

ChEMBL Bioactivity Layer

ChEMBL Database

- >19 million bioactivity records
- >2 million unique compounds
- Queried via REST API per target on demand

SQLite Cache

All API responses are cached locally with a 7-day TTL. Subsequent runs for the same target return in milliseconds.

Data Filtering Pipeline

1. Fetch IC50 measurements (human assays, nM units)
2. Validate SMILES with RDKit `MolFromSmiles`
3. Label: IC50 $\leq$ 1000 nM $\Rightarrow$ **active**
4. Stratified 80/20 train/test split

Rate Limiting

ChEMBL enforces rate limits. Retry logic with exponential backoff (3 attempts, 5 s base) prevents silent data loss.

QSAR Model Architecture

Molecular Featurisation

Morgan Fingerprints (radius = 2, 1024 bits)
Encode circular substructure neighbourhoods up to 2 bonds.

RDKit Descriptors (4 features concatenated):

- MolWt, LogP
- # H-bond donors (HBD)
- # H-bond acceptors (HBA)

Combined feature vector: $\mathbf{x} \in \mathbb{R}^{1028}$

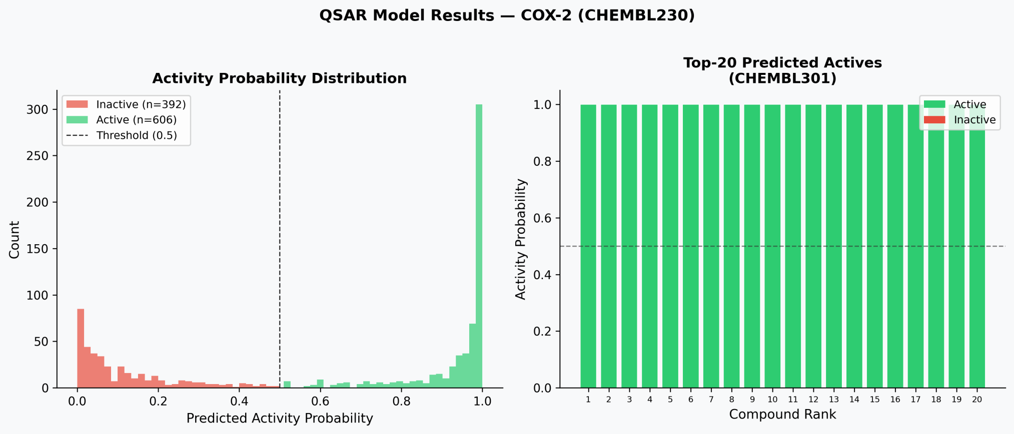
Random Forest Classifier

- 500 estimators
- 5-fold stratified cross-validation
- Reports ROC-AUC and F1 on held-out test set

Output

Per-compound predicted activity probability $p \in [0, 1]$. Top-20 compounds (p ranked descending) forwarded to docking.

QSAR Screening Results



Left: predicted activity probability distributions by class. *Right*: top-20 ranked compounds with active/inactive labels. CDK2 (CHEMBL301) — CV AUC = 0.943, Test F1 = 0.901.

AutoDock Vina Integration

Ligand Preparation

1. SMILES → 3D conformer (RDKit ETKDGv3)
2. Energy minimisation (MMFF94 force field)
3. Conversion to PDBQT (Open Babel)

Receptor Preparation

1. Structure fetch: AlphaFold → RCSB PDB
2. Conversion to PDBQT
3. Binding box: BioPython C_α centroid + 10 Å padding

Vina Scoring Function

Combines:

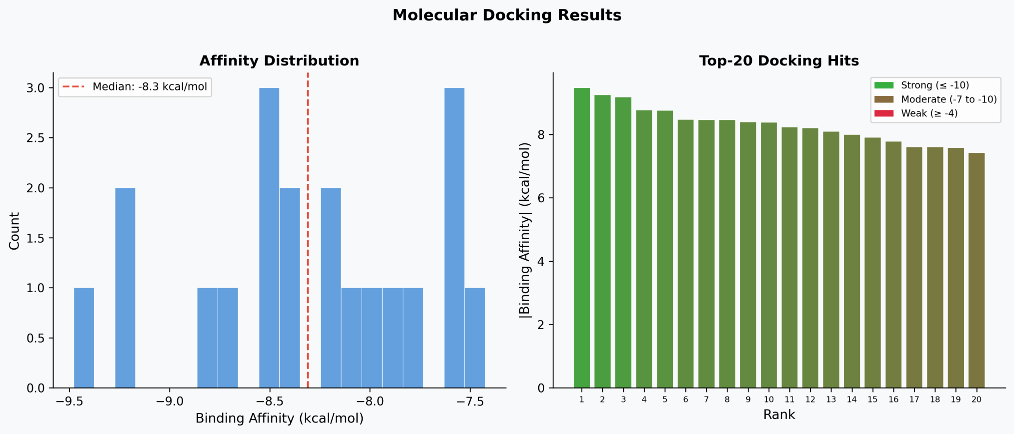
- Steric (Gaussian) terms
- Hydrogen-bond energy
- Hydrophobic contacts
- Torsional entropy penalty

Output: ΔG_{bind} in kcal/mol

Exhaustiveness

Set to 8 (default). Increase for larger binding pockets at the cost of runtime.

Docking Results



Left: full affinity distribution across screened candidates. *Right*: top-20 ranked hits coloured by binding strength (green $\leq -10\,\mathrm{kcal/mol}$, yellow moderate, red $\geq -4\,\mathrm{kcal/mol}$).

Case Study — Cyclin-Dependent Kinase 2 (CDK2)

Target Profile

ChEMBL ID	CHEMBL301
UniProt	P24941
Function	Cell cycle regulator (G1/S transition)
Disease link	Cancer (overexpression in many tumours)
Records	998 IC50 measurements retrieved

Docking — Top Hit

Best binding affinity: **−9.5 kcal/mol**

Affinity range of top-20: −9.5 to −7.4 kcal/mol

Compounds with $\Delta G \leq -9$ kcal/mol: **3/20**

QSAR Outcome

- CV ROC-AUC: **0.943** ± 0.012
- Test F1: **0.901**
- Training time: < 30 s

Total Pipeline Runtime

Stage	Time
ChEMBL fetch	∼110 s
QSAR training	<30 s
Docking (20 lig.)	∼5 min

APEX Web Platform

Backend — FastAPI

- `GET /targets/search?q=` — ChEMBL target lookup
- `POST /qsar/{target_id}` — train + predict
- `GET /structures/{uniprot_id}` — AlphaFold/PDB fetch
- `POST /docking/` — Vina run, ranked results

All routes are independently callable; each service is independently cacheable and testable.

Frontend — React

- Sidebar target search
- Real-time pipeline status log
- QSAR results table with activity probabilities
- Docking hit ranking with affinity heat colours
- Compound compare overlay

No Build Step

Single `index.html` with CDN React — open directly in a browser or serve with any static file server.

Summary & Future Work

What APEX Delivers

- Fully automated, target-agnostic screening pipeline
- ChEMBL bioactivity ingestion with local caching
- QSAR model (RF, AUC $>$ 0.94 on CDK2)
- AutoDock Vina docking for top-N candidates
- REST API + interactive web frontend

Planned Extensions

- Larger virtual compound libraries (Enamine REAL)
- Ensemble QSAR models (XGBoost, GNN)
- ADMET property filtering pre-docking
- 3D binding pose visualisation in-browser
- Multi-target comparative screening

APEX: an open, extensible platform that turns a target name into a ranked hit list — fully automatically.

Questions?

Source code available on request.

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